

Smart Monte Carlo for the LSE Dense Associative Memory: Budget Truncation and Validity Window

1 Motivation

Naive Metropolis–Hastings (MH) on the LSE Dense Associative Memory at load α stores $M = e^{\alpha N}$ random patterns; the per-step cost is $\mathcal{O}(M)$. The interesting limit for the AAAI paper is $\alpha \sim 1$, where $M = e^N$ — exponentially out of reach for $N \gtrsim 30$. This note describes a truncated (“smart”) MC scheme that retains a fixed budget $K \ll M$ of the most relevant patterns, derives the cutoff $\phi_{\text{cut}}(\alpha, N, K)$, and quantifies the temperature window in which the truncation remains faithful.

2 Setup

The LSE energy on the N -sphere $|\mathbf{x}|^2 = N$ is

$$H_{\text{LSE}}(\mathbf{x}) = -\frac{1}{\beta_{\text{net}}} \ln p(\mathbf{x}), \quad p(\mathbf{x}) = \sum_{\mu=1}^M e^{-\beta_{\text{net}} |\mathbf{x} - \boldsymbol{\xi}^\mu|^2/2}. \quad (1)$$

Writing $\phi_\mu = \mathbf{x} \cdot \boldsymbol{\xi}^\mu / N$ and $b \equiv \beta_{\text{net}} N$ gives the equivalent form

$$p(\mathbf{x}) = \sum_{\mu} e^{-b(1-\phi_\mu)}. \quad (2)$$

The MC dynamics samples $\mathbf{x}$ at temperature T from $\exp[-H_{\text{LSE}}/T]$. The parameters β_{net} and T are *independent*: β_{net} shapes the LSE kernel, T is the MC environment.

For patterns $\boldsymbol{\xi}^\mu$ drawn uniformly on $S^{N-1}(\sqrt{N})$, the marginal alignment with any fixed reference unit on the sphere is

$$\phi \sim f(\phi) = C_N (1 - \phi^2)^{(N-3)/2}, \quad \frac{1+\phi}{2} \sim \text{Beta}\left(\frac{N-1}{2}, \frac{N-1}{2}\right). \quad (3)$$

3 Budget truncation and the cutoff ϕ_{cut}

Near a stored pattern $\boldsymbol{\xi}^1$ the sum (2) is dominated by the champion $\mu = 1$ plus contributions from competitors with the largest ϕ_μ . We retain a budget of K live patterns, defined by the upper-tail condition $\phi_\mu > \phi_{\text{cut}}$, where

$$\boxed{\Pr(\phi > \phi_{\text{cut}}) = \frac{K}{M} = e^{\ln K - \alpha N}.} \quad (4)$$

Equivalently, in the Beta coordinate $x = (1 + \phi)/2$,

$$1 - I_{x_{\text{cut}}}\left(\frac{N-1}{2}, \frac{N-1}{2}\right) = \frac{K}{M}, \quad \phi_{\text{cut}} = 2x_{\text{cut}} - 1, \quad (5)$$

with $I_x(a, b)$ the regularised incomplete beta function. For $\log(K/M) \ll 0$ a stable evaluation uses the leading asymptotic $a \ln u \simeq \ln(K/M) + \ln a + \ln B(a, a)$ with $u = 1 - x_{\text{cut}}$, refined by Newton iteration in $\ln u$.

The remaining $M - K$ patterns form a “passive sea”. On the sphere, by rotation invariance,

$$\sum_{\mu \notin \text{live}} e^{-b(1-\phi_\mu)} \stackrel{N \rightarrow \infty}{\equiv} C(\alpha, N, K) + \mathcal{O}(\sqrt{M-K}), \quad (6)$$

with mean $C = (M-K) \mathbb{E}_f[e^{-b(1-\phi)}]$ computed from (3). By the law of large numbers over $(M-K)$ i.i.d. weak contributions, the fluctuations of the passive sum about its mean are subdominant; we replace it by the constant C in the simplified dynamics:

$$\tilde{p}(\mathbf{x}) \equiv \sum_{\mu \in \text{live}} e^{-b(1-\phi_\mu(\mathbf{x}))} + C. \quad (7)$$

Crucially, C enters *inside* the logarithm in $\tilde{H} = -\beta_{\text{net}}^{-1} \ln \tilde{p}$ — it is *not* an additive shift of the energy. Where the live sum exceeds C , the dynamics is the truncated LSE; where it falls below C , the energy flattens into a noise plateau.

4 Bare validity bound

The smart-MC approximation is faithful while the MH chain in equilibrium around ξ^1 keeps the champion’s alignment ϕ_1 above ϕ_{cut} . The single-basin equilibrium gives

$$\phi_{\text{eq}}(T) = \frac{1}{2} \left[-T + \sqrt{T^2 + 4} \right] \quad (8)$$

(independent of β_{net} , since the $1/\beta_{\text{net}}$ in (1) cancels for a single dominant pattern). Demanding $\phi_{\text{eq}}(T) \geq \phi_{\text{cut}}$ inverts to

$$\boxed{T_{\text{max}}(\alpha, N, K) = \frac{1 - \phi_{\text{cut}}^2}{\phi_{\text{cut}}}.} \quad (9)$$

Above T_{max} the equilibrium drives $\mathbf{x}$ into territory where no live pattern has appreciable overlap with it, and the simulation collapses onto the artificial noise floor C .

5 Thermal fluctuation correction

The bare bound (9) ignores Gaussian fluctuations of ϕ about ϕ_{eq} . Working from

$$P(\phi) \propto \exp \left(N \left[\phi/T + \frac{1}{2} \ln(1 - \phi^2) \right] \right), \quad (10)$$

the saddle is (8) and the second derivative of the per-spin exponent is

$$f''_{\text{ps}}(\phi_{\text{eq}}) = \frac{1 + \phi_{\text{eq}}^2}{(1 - \phi_{\text{eq}}^2)^2} \cdot T. \quad (11)$$

The variance of ϕ around ϕ_{eq} is then

$$\boxed{\sigma_\phi^2 = \frac{T}{N f''_{\text{ps}}(\phi_{\text{eq}})} = \frac{(1 - \phi_{\text{eq}}^2)^2}{N (1 + \phi_{\text{eq}}^2)}.} \quad (12)$$

Two points to flag. (i) The explicit T cancels in (12). The dependence on T enters only through $\phi_{\text{eq}}(T)$. (ii) As $T \rightarrow 0$, $\phi_{\text{eq}} \rightarrow 1$ and $\sigma_\phi \rightarrow 0$ as $(1 - \phi_{\text{eq}}^2)/\sqrt{N}$. This is *not* a generic “small- T implies small fluctuations” statement; rather, the geometric-entropy curvature $\propto 1/(1 - \phi^2)^2$ diverges as the saddle approaches the sphere wall $\phi = 1$, and that diverging stiffness suppresses fluctuations even though T itself cancels.

A naive equipartition estimate $\sigma \sim \sqrt{T/N}$ — appropriate for a free particle — overestimates σ_ϕ by a factor $\sqrt{N(1 + \phi_{\text{eq}}^2)/(1 - \phi_{\text{eq}}^2)}$, which is large in the low- T regime that controls retrieval.

A safety margin $k\sigma_\phi$ ($k \sim 3$) yields the refined bound

$$T_{\text{safe}}(\alpha, N, K; k) : \quad \phi_{\text{eq}}(T_{\text{safe}}) - k\sigma_\phi(\phi_{\text{eq}}(T_{\text{safe}}), N) = \phi_{\text{cut}}, \quad (13)$$

solved by bisection. With σ_ϕ from (12), this correction is small in the regime relevant to the LSE retrieval boundary.

6 Physical retrieval boundary $T_{\text{crit}}(\alpha)$

The exact LSE retrieval phase boundary follows from setting the retrieval free-energy density equal to the noise floor (cf. LSE_gaussian.tex):

$$f_{\text{ret}}(T) = 1 - \phi_{\text{eq}}(T) - \frac{T}{2} \ln(1 - \phi_{\text{eq}}^2(T)), \quad \alpha_c^{\text{E}}(T) = -\frac{1}{2} \ln(1 - (1 - f_{\text{ret}})^2). \quad (14)$$

Inverting $\alpha_c^{\text{E}}(T) = \alpha$ gives the temperature $T_{\text{crit}}(\alpha)$ at which retrieval fails at load α . The smart-MC simulation is meaningful for measuring this boundary provided

$$\boxed{T_{\text{max}}(\alpha, N, K) > T_{\text{crit}}(\alpha) \quad \text{for every } \alpha \text{ we sweep.}} \quad (15)$$

7 Validity diagram and design choices

Monotonicity. At fixed K , increasing N moves ϕ_{cut} closer to 1 (the budget covers a thinner tail), shrinking T_{max} via (9). The shrinkage is bounded: as $N \rightarrow \infty$, $\phi_{\text{cut}} \rightarrow \phi_{\text{max}}(\alpha) = \sqrt{1 - e^{-2\alpha}}$, so T_{max} approaches an N -independent floor $(1 - \phi_{\text{max}}^2)/\phi_{\text{max}} = e^{-2\alpha}/\sqrt{1 - e^{-2\alpha}}$ at the same α . For $\alpha=1$ the floor is $T_{\text{max}} \rightarrow 0.142$, while $T_{\text{crit}}(1) \approx 0.031$; the validity inequality (15) survives the large- N limit by a factor ~ 5 .

Concrete operating points for the AAAI sweep. Table 1 (computed in `plot_smart_MC_validity.jl`) lists $T_{\text{max}}(\alpha, N, K)$ relative to $T_{\text{crit}}(\alpha)$ for the canonical configurations.

Reading the design space. For $\alpha \leq 1.2$, $N=100$ with $K=10^4$ already provides a $\sim 6\times$ cushion above T_{crit} . To extend cleanly to $\alpha=1.5$, $N=200$ with $K=10^4$ keeps a $\sim 6\times$ cushion. Pushing K from 10^4 to $e^{12} \approx 1.6 \times 10^5$ only moves T_{max} by a few percent — the dominant constraint is on K rather than on N for $\alpha \leq 1$, with the roles partially reversing only at high α .

8 Two thresholds in α

The analysis of the smart-MC scheme uncovers *two* distinct α -thresholds, easy to confuse but with different meanings.

config	$\alpha=0.5$	$\alpha=0.7$	$\alpha=1.0$	$\alpha=1.2$	$\alpha=1.5$
$T_{\text{crit}}(\alpha)$ (theory)	0.135	0.073	0.031	0.019	0.009
$N=50, K=10^4$	0.896	0.497	0.239	0.152	0.080
$N=100, K=10^4$	0.636	0.375	0.187	0.121	0.064
$N=100, K=e^{12}$	0.690	0.402	0.199	0.128	0.068
$N=200, K=10^4$	0.543	0.327	0.165	0.108	0.057
$N=500, K=10^4$	0.494	0.301	0.153	0.100	0.054

Table 1: Bare validity temperature $T_{\text{max}}(\alpha, N, K)$ vs. the physical retrieval temperature $T_{\text{crit}}(\alpha)$ for several smart-MC operating points (LSE, $\alpha \in [0.5, 1.5]$). All listed configurations satisfy (15) across the table.

Closed forms. The mean weight $\langle w \rangle = \mathbb{E}_f[e^{-\beta_{\text{net}} N(1-\phi)}]$ is dominated, at large N , by a saddle at $\phi^* = (\sqrt{5} - 1)/2 = 1/\varphi_g$ (inverse golden ratio):

$$\langle w \rangle \approx e^{-\lambda_w N}, \quad \lambda_w = 2 - \varphi_g + \frac{1}{2} \ln \varphi_g \approx 0.6226, \quad (16)$$

with $\varphi_g = (1 + \sqrt{5})/2$. The second moment $\langle w^2 \rangle$ has its own saddle at $\phi_2^* = (\sqrt{17} - 1)/4$, giving

$$\langle w^2 \rangle \approx e^{-\lambda_{w^2} N}, \quad \lambda_{w^2} \approx 0.909. \quad (17)$$

Define the relative-variance exponent

$$\lambda_{\text{var}} = 2\lambda_w - \lambda_{w^2} \approx 0.336. \quad (18)$$

Threshold #1 — self-averaging ($\alpha > \lambda_{\text{var}} \approx 0.34$).

$$\frac{\sigma_S}{C} = \frac{1}{\sqrt{M-K}} \sqrt{\frac{\langle w^2 \rangle}{\langle w \rangle^2}} \approx \exp\left(\frac{1}{2}(\lambda_{\text{var}} - \alpha) N\right), \quad (19)$$

which is $\ll 1$ iff $\alpha > \lambda_{\text{var}}$. **This is a hard validity bound on the mean-replacement.** It cannot be repaired by adjusting K : σ_S/C scales as $1/\sqrt{M-K}$, so larger K only worsens self-averaging. Below λ_{var} , the heavy-tailed weight distribution is intrinsically non-self-averaging at any (N, K) , and one must fall back on honest MC.

Threshold #2 — floor-vs-live crossover ($\alpha \approx \lambda_w \approx 0.62$).

$$C(\alpha, N) \approx e^{(\alpha - \lambda_w) N}, \quad (20)$$

so C crosses unity at $\alpha = \lambda_w$. For $\alpha < \lambda_w$, $C \ll p_{\text{live}}$ and the dynamics is dominated by the live patterns:

$$H \approx -\frac{1}{\beta_{\text{net}}} \ln p_{\text{live}} + \mathcal{O}(C/p_{\text{live}}). \quad (21)$$

For $\alpha > \lambda_w$, $C \gg p_{\text{live}}$ and the log expands the other way:

$$H \approx -\frac{1}{\beta_{\text{net}}} \left[\ln C + \frac{p_{\text{live}}}{C} \right], \quad (22)$$

i.e. a constant shift plus the live signal rescaled by $1/C$. **This is not a validity threshold:** the simulator evaluates $H = -\beta_{\text{net}}^{-1} \ln(C + p_{\text{live}})$ exactly via max-shift LSE, and the basin shape in ϕ -space is preserved on both sides. Only the analytical reasoning about which term sets the energy scale differs.

Where the AAAI sweep sits. The simulator covers $\alpha \in [0.40, 1.00]$ at $N = 500$, $K = 10^4$, $\beta_{\text{net}} = 1$.

Table 2: Regime map of the AAAI sweep.

α window	self-averaging	dominant LSE term	σ_S/C at $N = 500$
0.40–0.62	yes (well above λ_{var})	p_{live}	$\leq 10^{-15}$
0.62–1.00	yes	C	$\rightarrow 10^{-70}$

The MH validity condition $\phi_{\text{eq}}(T) > \phi_{\text{cut}}$, derived in Sec. 4, is geometric and applies in both regimes — the chain escapes the basin when thermal fluctuations of ϕ_1 reach the nearest live competitor’s overlap, independently of whether C or p_{live} dominates the energy magnitude.

9 Simulation grids

α -grid: $\alpha \in \{0.40, 0.45, 0.50, \dots, 1.00\}$ — 13 values, step 0.05.

T -grid (per α): 30 linearly-spaced points from $T_{\text{min}} = 0.005$ up to $0.9T_{\text{max}}(\alpha, N, K)$, with $T_{\text{max}} = (1 - \phi_{\text{cut}}^2)/\phi_{\text{cut}}$ from (9). The grid is therefore a *wedge* narrowing at large α , with resolution naturally densest where $T_{\text{crit}}(\alpha)$ lives.

Table 3: T-grid range per α at $N = 500$, $K = 10^4$.

α	ϕ_{cut}	T_{max}	top of grid	T_{crit} (physical)
0.40	0.838	0.61	0.55	≈ 0.20
0.50	0.873	0.43	0.39	0.135
0.70	0.918	0.25	0.22	0.073
1.00	0.951	0.15	0.14	0.031

Disorder samples. n_{disorder} ramps from 128 at $\alpha = 0.40$ down to 32 at $\alpha = 1.00$ (cheap at high α , expensive at low α — but signal cleaner there).

Hardware. A single NVIDIA A6000 (48 GB) accommodates the full sweep at $N = 500$, $K = 10^4$, $n_{\text{dis,max}} = 128$ in one chunk (peak memory ~ 33 GB out of a 40 GB budget). Expected wall time ~ 30 minutes.

10 Consistency check with honest Monte Carlo

The cleanest validation runs at moderate N where honest (untruncated) MC is still tractable. We propose:

1. Fix $N = 15$ and run honest MC across $\alpha \in \{0.2, 0.5, 0.8, 1.0\}$ with $M = e^{\alpha N}$. This is the ground truth.
2. Run smart MC at the same (N, α) with K swept from $K = M$ down to $K = 100$, using the truncated dynamics (7) with C from (6).

3. Plot $\langle\phi\rangle(K)$ at fixed T . The curve must be flat (within MC error) at $K=M$ and only bend systematically as K enters the regime where (15) fails.
4. The systematic bias is upward in $\langle\phi\rangle$: smart MC overestimates retrieval because the passive sea's x -dependent fluctuations are replaced by their mean.

A second sanity check is trivial: for $\alpha < 0.2$ the budget $K = 10^4$ already exceeds M at $N=30$ –50, so smart MC reduces to honest MC by construction.

11 Open question: why does $\alpha_c^E(T)$ not depend on β_{net} ?

Observation. The exact LSE retrieval boundary $\alpha_c^E(T)$ derived in `LSE.gaussian.tex` has no β_{net} in it. This is at first puzzling because β_{net} visibly controls the LSE kernel sharpness in (1) and, intuitively, should set how distinct basins are.

Working hypothesis. The single-pattern retrieval analysis collapses the LSE *onto its dominant term* near a stored pattern ξ^1 :

$$H_{\text{LSE}}(\mathbf{x}) = -\frac{1}{\beta_{\text{net}}} \ln p(\mathbf{x}) \xrightarrow{\mathbf{x} \rightarrow \xi^1} -\frac{1}{\beta_{\text{net}}} \ln[e^{-\beta_{\text{net}} N(1-\phi_1)}] = N(1-\phi_1), \quad (23)$$

so the $1/\beta_{\text{net}}$ in front of $\ln p$ exactly cancels the β_{net} in the exponent. The free-energy density $f_{\text{ret}}(T)$ depends only on $(1-\phi)$ and the geometric entropy $s(\phi)$. The same cancellation occurs on the noise-floor side: the spurious branch energy is $1-\varphi_{\text{max}}$, again with no β_{net} .

Where β_{net} then survives. It controls only the *sub-leading* log-sum-exp correction that mixes the champion with the next-best competitor. Concretely, near ξ^1 ,

$$H = N(1-\phi_1) - \frac{1}{\beta_{\text{net}}} \ln \left[1 + \sum_{\mu \geq 2} e^{-\beta_{\text{net}} N(\phi_1 - \phi_\mu)} \right]. \quad (24)$$

For $\beta_{\text{net}} N(\phi_1 - \phi_\mu) \gg 1$ this correction is exponentially suppressed and the dynamics is governed entirely by the champion. The retrieval/no-retrieval transition is then set by the *location* of the spurious maximum φ_{max} , which is a property of the pattern distribution and is itself independent of β_{net} .

To verify in the AAAI paper. Run the smart-MC simulator (Sec. 7) at several values of β_{net} on the same (α, T, N, K) grid and check that the measured boundary collapses onto one curve. If it does not collapse cleanly, the dependence we predict to be sub-leading is in fact load-bearing in a finite- N regime, and the analytic boundary $\alpha_c^E(T)$ should be re-derived with an explicit β_{net} -dependent finite-temperature competition term.

12 Summary

Smart MC for the LSE memory keeps the K patterns with the largest overlap with a reference, replaces the remaining $M-K$ patterns by their mean contribution C inside the log, and uses the truncated $\tilde{p}$ for MH dynamics. The scheme is faithful provided $T_{\text{max}}(\alpha, N, K) = (1 - \phi_{\text{cut}}^2)/\phi_{\text{cut}} > T_{\text{crit}}(\alpha)$, with ϕ_{cut} determined by the upper-tail condition (4). Thermal corrections to this bound, computed from the proper saddle curvature, are small and do not change the picture: the validity inequality survives the large- N limit by a factor ~ 5 at $\alpha = 1$, leaving ample design room for the AAAI sweep.